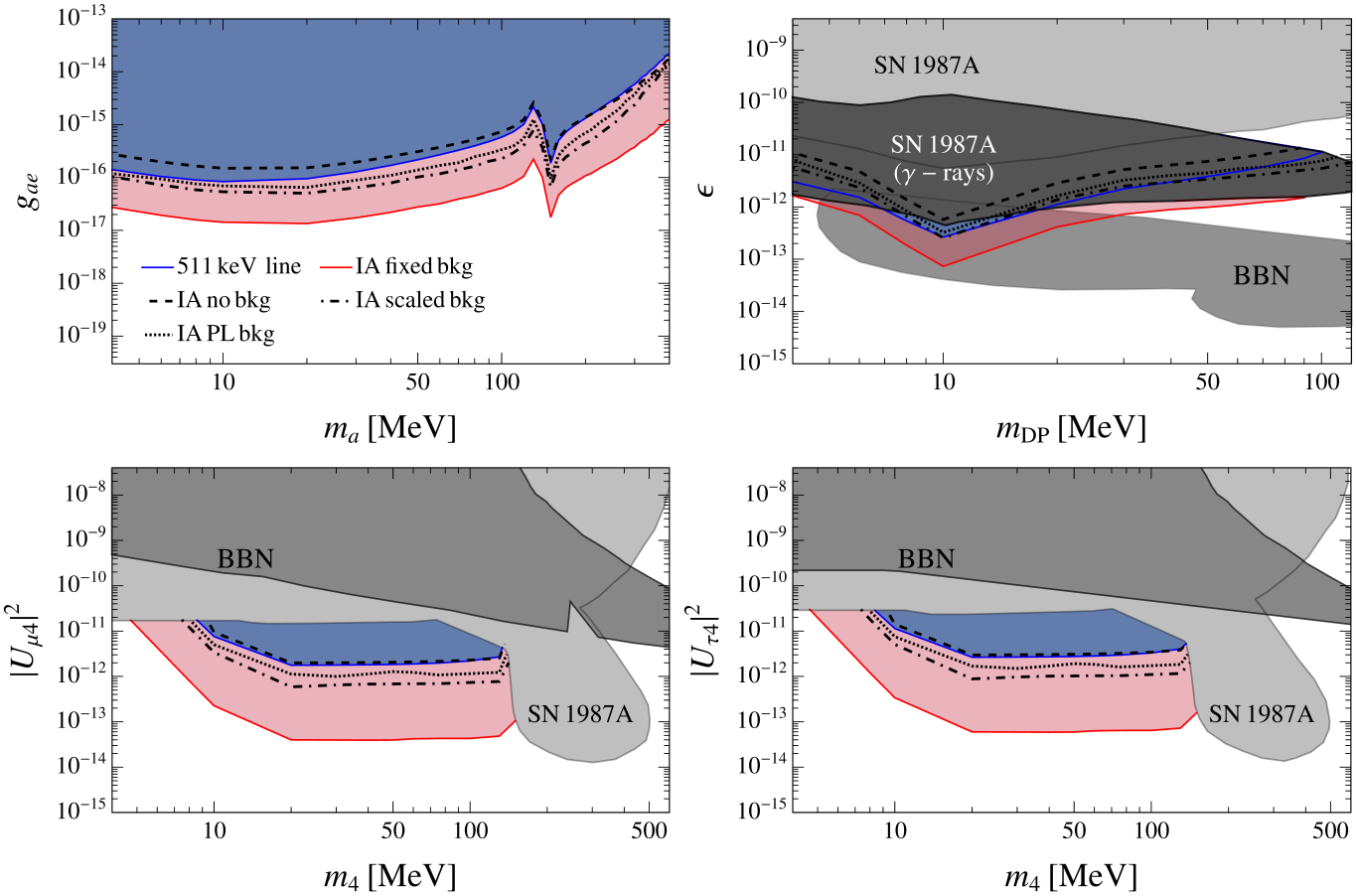

FIG. 9. Upper limits, at 2σ confidence level, for the case of ALPs coupled to nucleons and electrons with $g_{ap} = 2 \times 10^{-11}$ (upper left panel), DPs (upper right panel), and sterile neutrinos (lower panels). We show the limits obtained in the different analyses performed: Including our background model ("fixed bkg"), with the same background model but letting it's normalization be free in the fit ("scaled bkg"), including a background described as a simple power-law ("PL bkg") and, the most conservative case, without accounting for any background ("no bkg"). For completeness, we also include the limit obtained from the analysis of the 511 keV emission. SN 1987A and BBN constraints on the DP parameter space have been introduced in Refs. [17, 86, 130] and Refs. [131–133], respectively. Analogously, other bounds on the sterile neutrino parameter space are placed by observations of SN 1987A [15] and cosmological constraints [134–136].

signals needed on top of the background, to reproduce COMPTEL data. In all the background configurations and FIP models, we found that the IA signals do not lead to a significant improvement of the fit to COMPTEL data, meaning that the background suffices to explain these observations. In the case of the adopted reference background model, we find that the best-fit normalization is always within 10% of the original prediction. As an example, we illustrate, in the form of a triangle plot, the posterior distributions obtained in the case of the signal from a 30 MeV sterile neutrino assuming a power-law background, in Fig. 8. See more details in the label of the figure. In this example, one can see that the best-fit coupling of the signal is consistent with no coupling, but the upper limit (at 2σ confidence) still allows non-negligible couplings.

The right panel of Fig. 7 shows the limits obtained for ALPs coupling to baryons and electrons in the conservative case (i.e. including no background – "No Bkg") and in the most optimistic case (obtained with our reference background – "Fixed Bkg"). The bands correspond to the uncertainties associated with the propagation parameters, as discussed previously. Referring to the discussion in Sec. II, we warn the reader about the behavior of the constraints around the range $m_a \simeq 135\,\text{MeV}$ and $m_a \simeq 147\,\text{MeV}$, where resonant effects on the induced ALP-photon coupling could affect the results of the analysis. In particular, around $m_a \simeq 135\,\text{MeV}$ positron injection is suppressed since all the ALPs tend to decay in photons. Thus, our constraints relax. Conversely, around $m_a \simeq 147\,\text{MeV}$ electronic decays are enhanced by the suppression of the induced ALP-photon coupling and our limits get more stringent. As we see, the difference in the limits obtained when accounting for the background